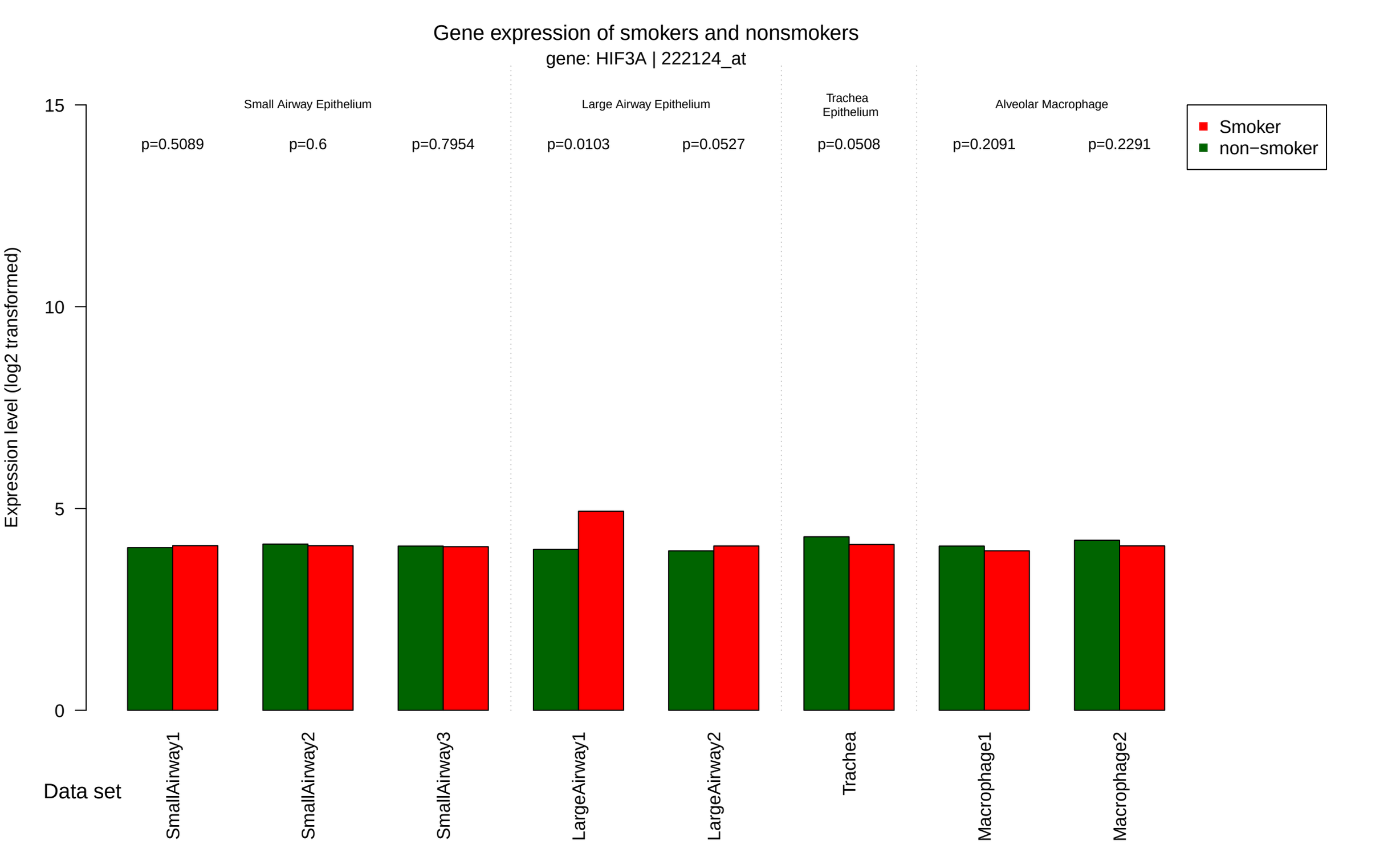
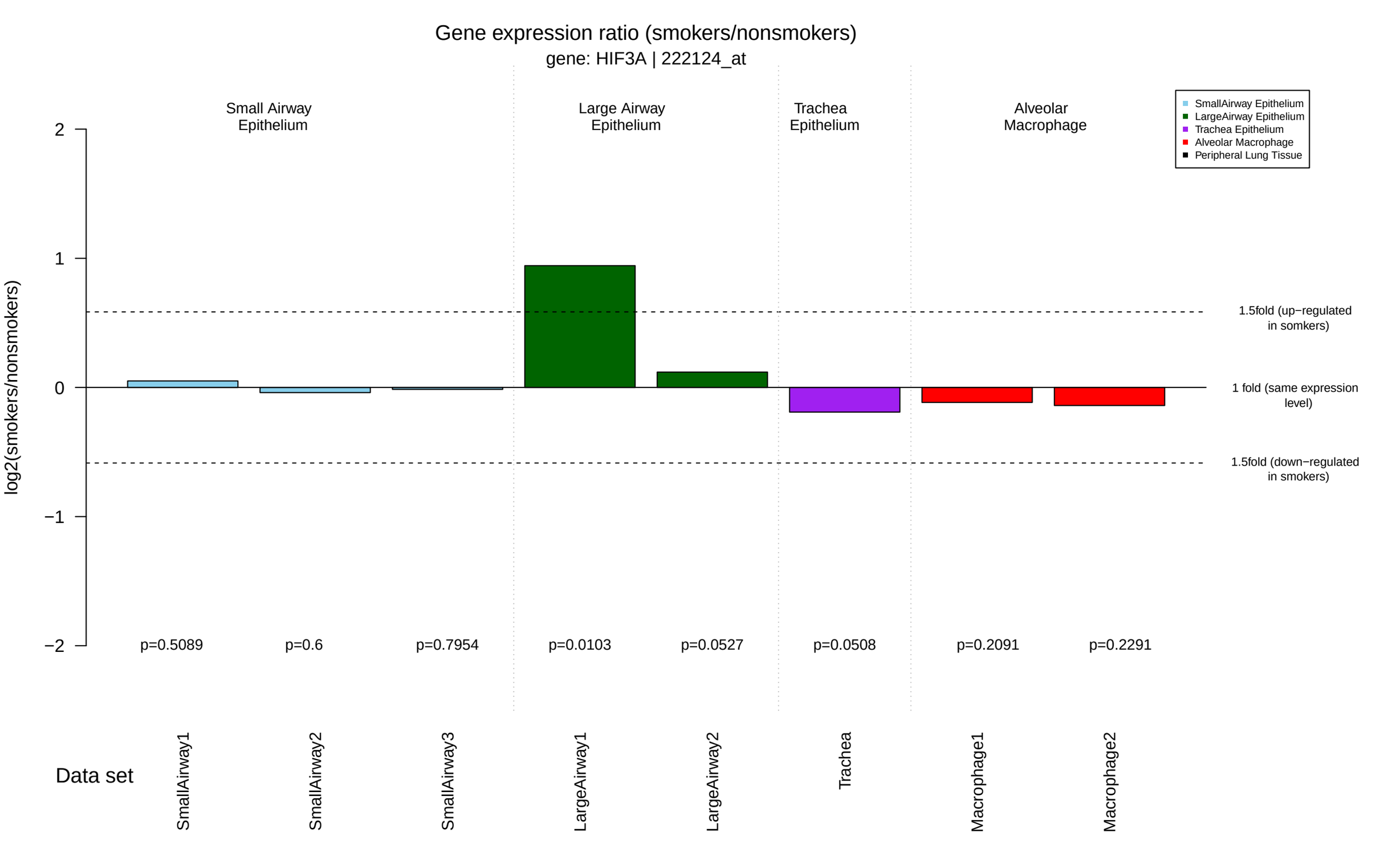
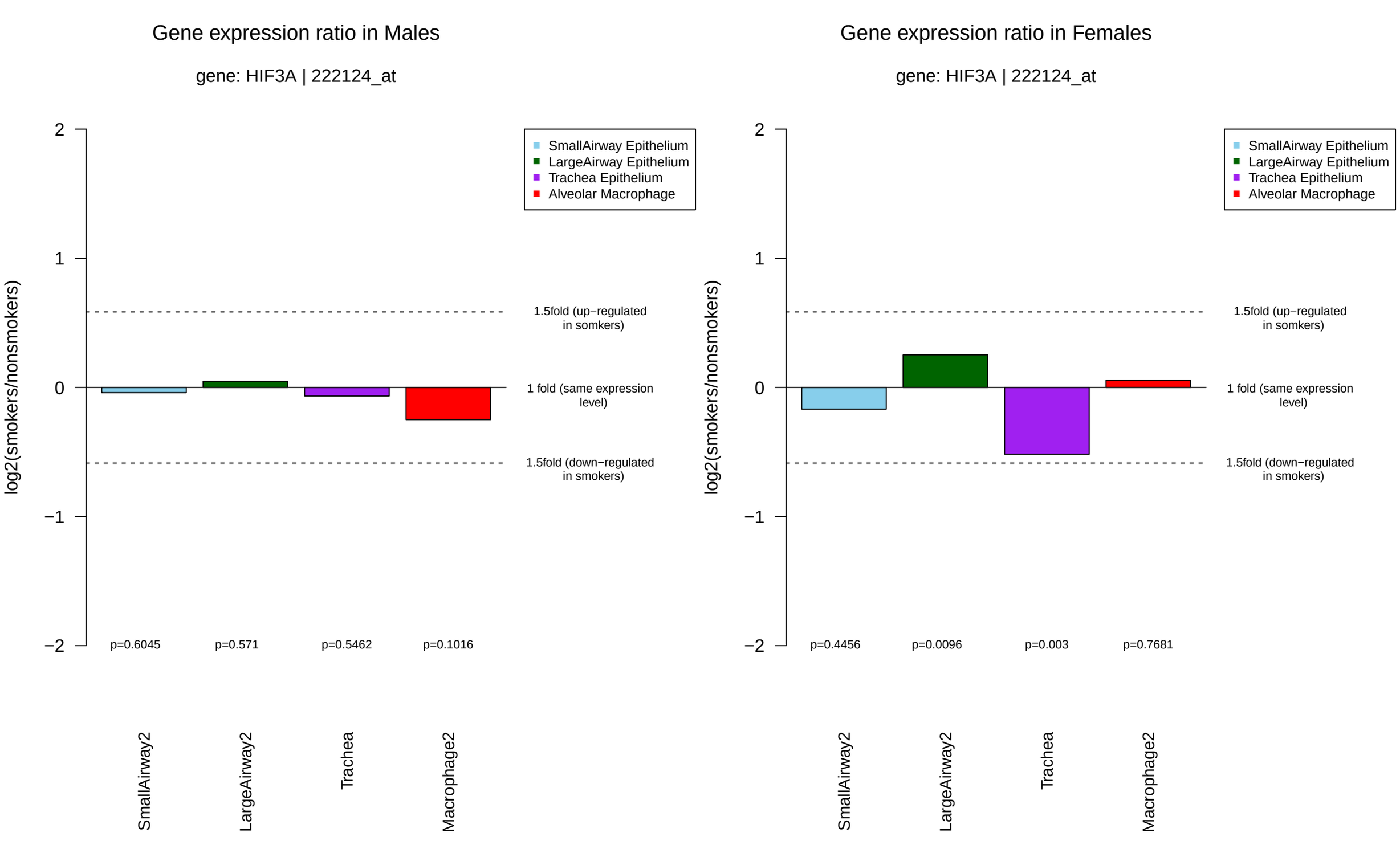
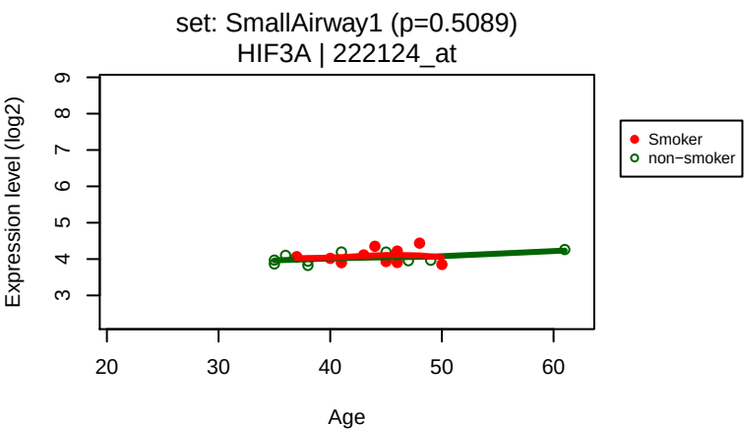
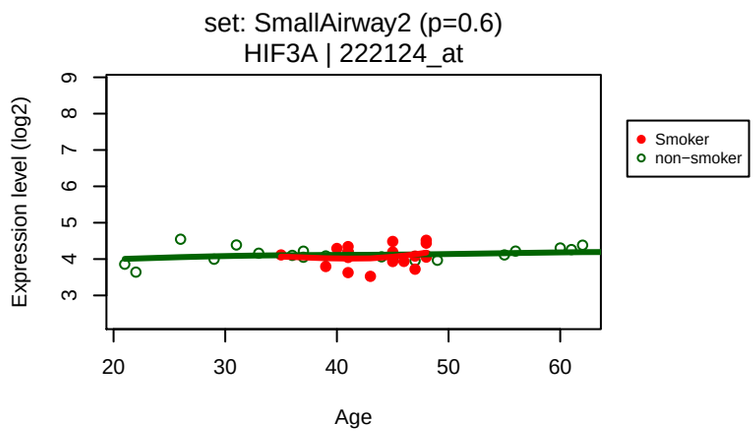
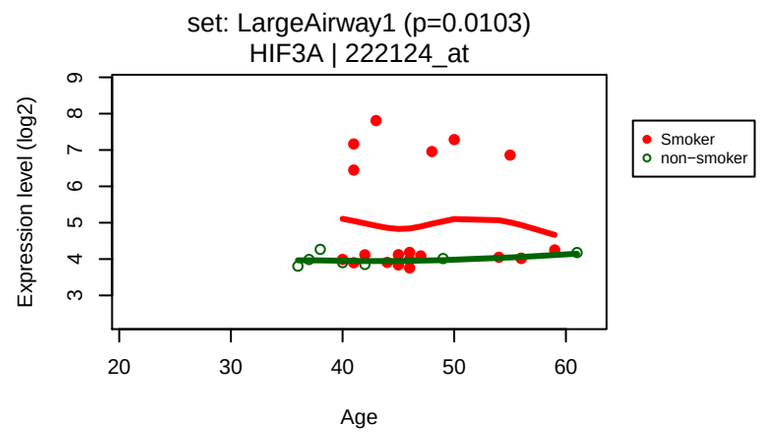

Gene expression of smokers and nonsmokers
gene: HIF3A | 222124_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5089
p=0.6
p=0.7954
p=0.0103
p=0.0527
p=0.0508
p=0.2091
p=0.2291
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

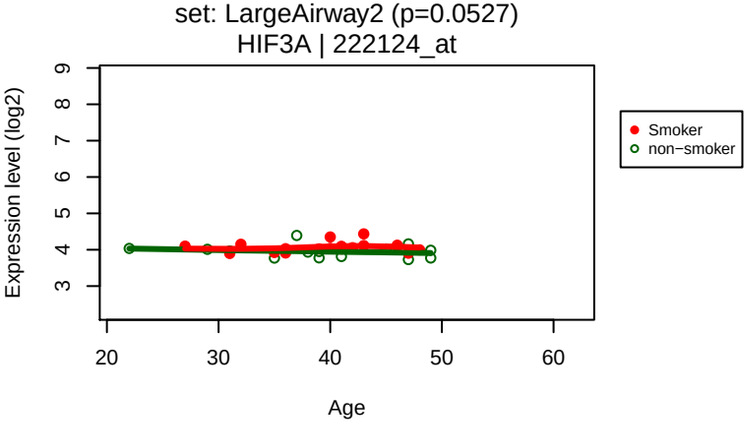

Gene expression ratio (smokers/nonsmokers)
gene: HIF3A | 222124_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5089
p=0.6
p=0.7954
p=0.0103
p=0.0527
p=0.0508
p=0.2091
p=0.2291
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

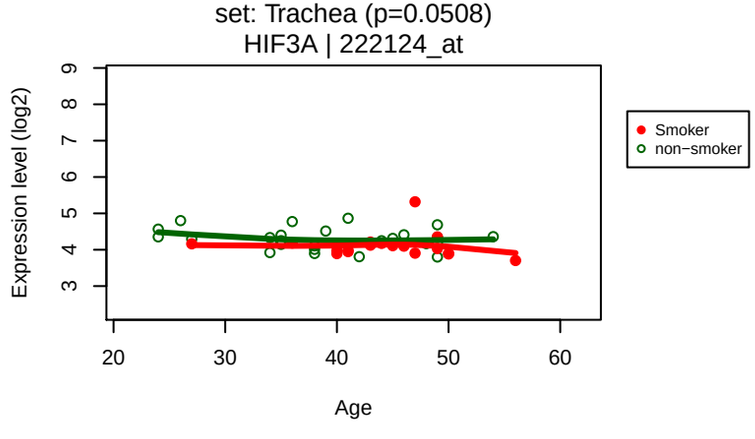

Gene expression ratio in Males
gene: HIF3A | 222124_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6045
p=0.571
p=0.5462
p=0.1016
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIF3A | 222124_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4456
p=0.0096
p=0.003
p=0.7681
SmallAirway2
LargeAirway2
Trachea
Macrophage2

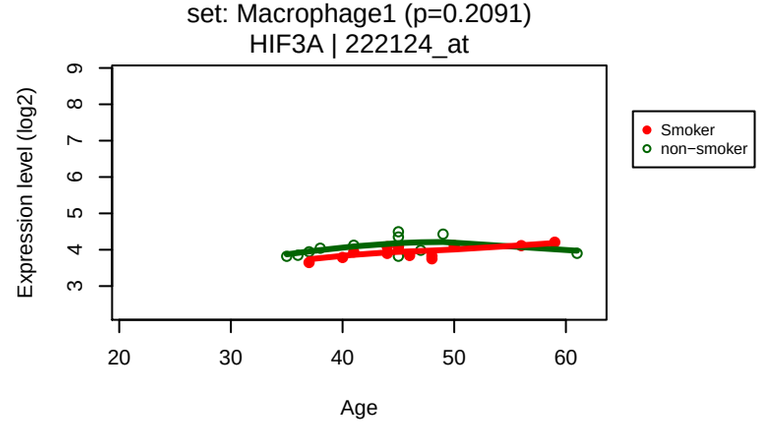

set: SmallAirway1 (p=0.5089)
HIF3A | 222124_at
Expression level (log2)
Age
Smoker
non−smoker

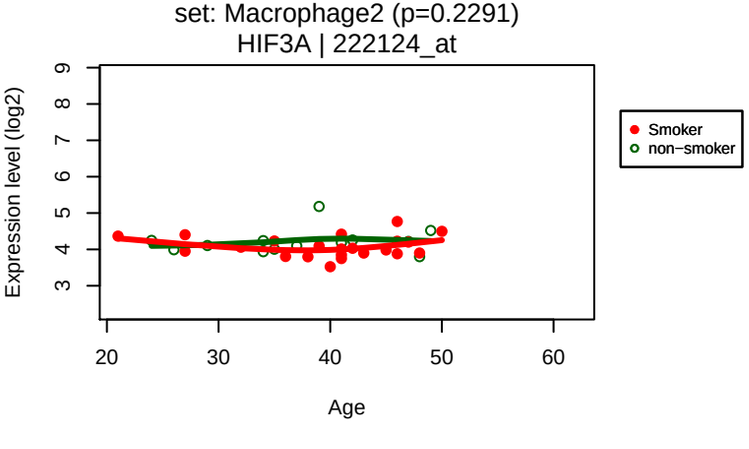

set: SmallAirway2 (p=0.6)
HIF3A | 222124_at
Expression level (log2)
Age
Smoker
non−smoker

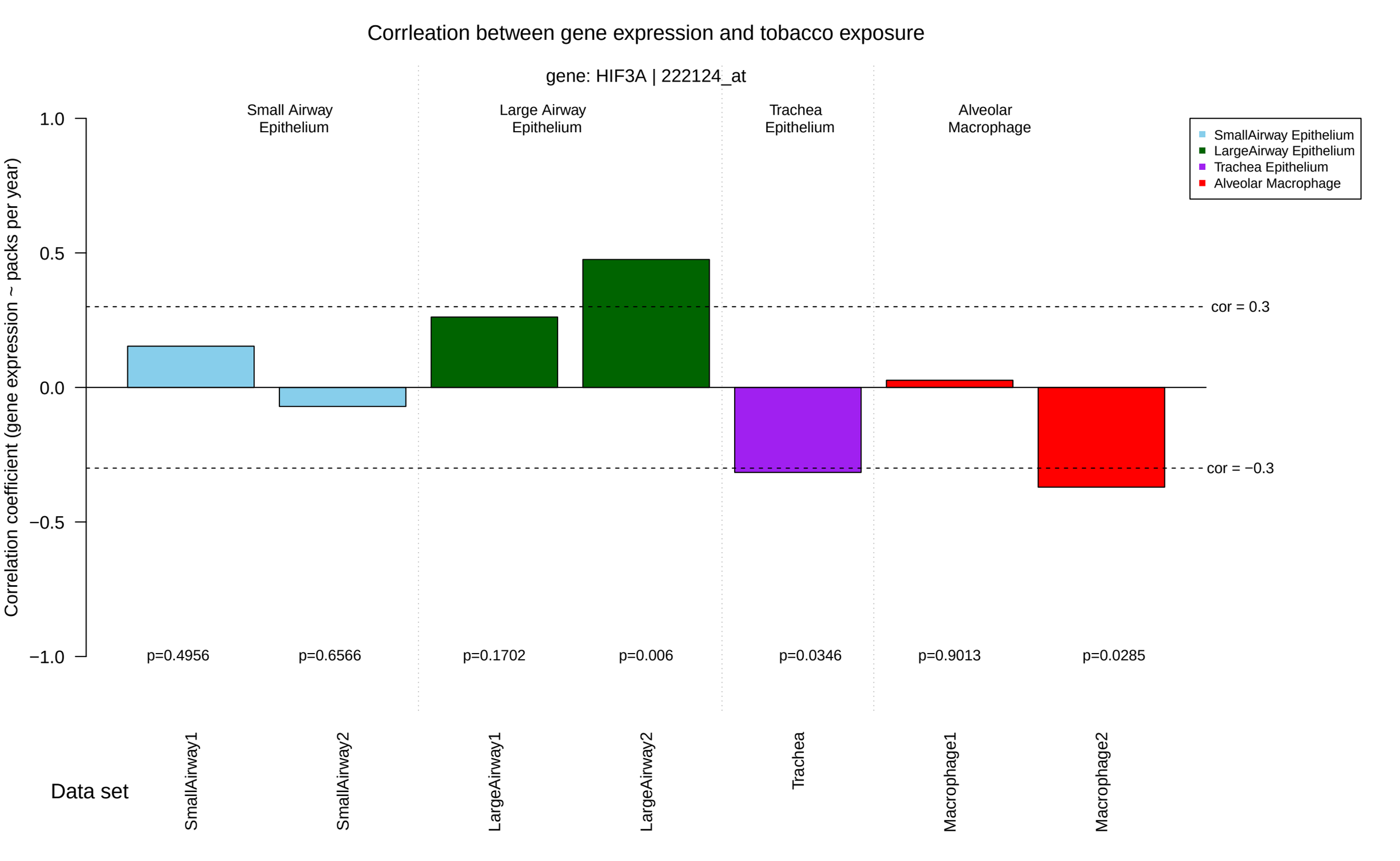

set: LargeAirway1 (p=0.0103)
HIF3A | 222124_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0527)
HIF3A | 222124_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0508)
HIF3A | 222124_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2091)
HIF3A | 222124_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2291)
HIF3A | 222124_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIF3A | 222124_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4956
p=0.6566
p=0.1702
p=0.006
p=0.0346
p=0.9013
p=0.0285
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2